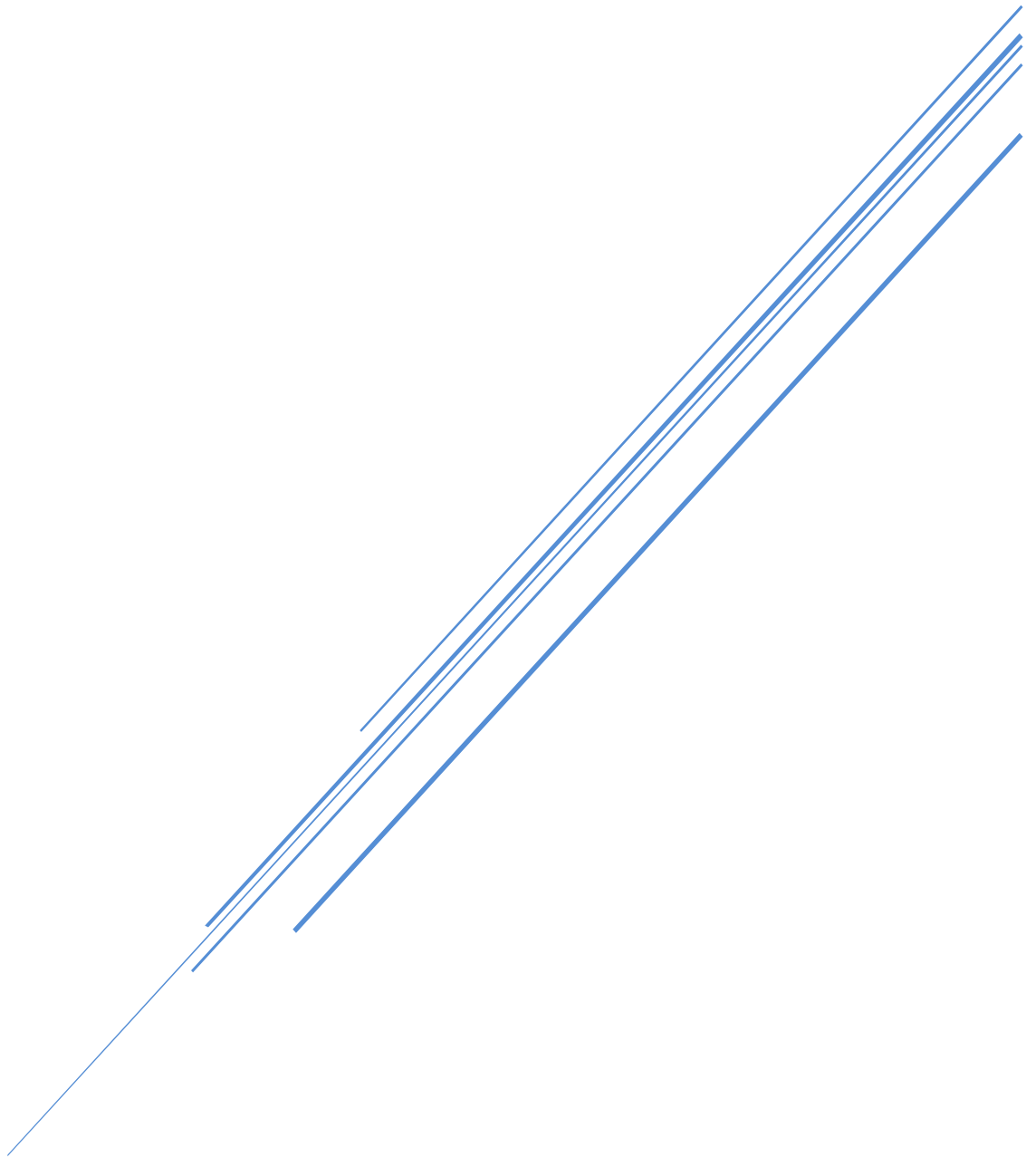


Enterprise Network Design and Implementation Using Huawei Datacom Technologies



Core Link Technologies
Full Documentation

3. Physical & Interface Mapping

A mapping of all inter-device connections including router-to-router, core-to-aggregation, aggregation-to-access, and access-to-AP infrastructure.

All EtherChannel (Eth-Trunk) bundles are documented, including port assignments and uplink distribution.

HQ Site

HQ Core

Device	Interface	Connected To	Interface	Notes
HQ-R1	G0/0/0	HQ-1	G0/0/1	
HQ-R1	G0/0/1	ISP1	G0/0/0	
HQ-1	G0/0/24	Core-SW1	G0/0/22	
HQ-1	G0/0/23	Core-SW2	G0/0/22	
Core-SW1	G0/0/22	HQ-1	G0/0/24	
Core-SW1	G0/0/23,24	Core-SW2	G0/0/23,24	EtherChannel
Core-SW1	G0/0/19,20	Agg-SW1	G0/0/23,24	EtherChannel
Core-SW1	G0/0/17,18	Agg-SW2	G0/0/23,24	EtherChannel
Core-SW1	G0/0/15,16	Agg-SW3	G0/0/23,24	EtherChannel
Core-SW2	G0/0/23,24	Core-SW1	G0/0/23,24	EtherChannel
Core-SW2	G0/0/19,20	Agg-SW3	G0/0/21,22	EtherChannel
Core-SW2	G0/0/17,18	Agg-SW2	G0/0/21,22	EtherChannel
Core-SW2	G0/0/15,16	Agg-SW1	G0/0/21,22	EtherChannel

HQ Aggregation & Access

Device	Interface	Connected To	Interface	Notes
Agg-SW1	G0/0/23,24	Core-SW1	G0/0/19,20	EtherChannel
Agg-SW1	G0/0/21,22	Core-SW2	G0/0/15,16	EtherChannel
Agg-SW2	G0/0/23,24	Core-SW1	G0/0/17,18	EtherChannel
Agg-SW2	G0/0/21,22	Core-SW2	G0/0/17,18	EtherChannel
Agg-SW3	G0/0/23,24	Core-SW1	G0/0/15,16	EtherChannel
Agg-SW3	G0/0/21,22	Core-SW2	G0/0/19,20	EtherChannel
Agg-SW3	G0/0/19,20	AC1	G0/0/7,8	EtherChannel
Agg-SW3	G0/0/3	VoIP Server	Ethernet0/0/0	
Agg-SW3	G0/0/2	NVR Server	Ethernet0/0/0	
Agg-SW3	G0/0/1	File Storage	Ethernet0/0/0	
AC1	G0/0/7,8	Agg-SW3	G0/0/19,20	EtherChannel
AC1	G0/0/2	AP3	G0/0/4	
AC1	G0/0/4	AP1	G0/0/4	
AC1	G0/0/6	AP2	G0/0/4	



Branch Site (BR1)

Device	Interface	Connected To	Interface	Notes
BR1-R1	G0/0/1	ISP2	G0/0/1	
BR1-R1	G0/0/0	BR-1	G0/0/1	
BR-1	G0/0/23	BR1-CoreSwitch1	G0/0/16	
BR-1	G0/0/24	BR1-CoreSwitch2	G0/0/16	
CoreSwitch1	G0/0/21,22	CoreSwitch2	G0/0/21,22	EtherChannel
CoreSwitch1	G0/0/17,18	AG-SW-1	G0/0/23,24	EtherChannel
CoreSwitch1	G0/0/19,20	AG-SW-2	G0/0/21,22	EtherChannel
CoreSwitch2	G0/0/21,22	CoreSwitch1	G0/0/21,22	EtherChannel
CoreSwitch2	G0/0/17,18	AG-SW-1	G0/0/21,22	EtherChannel
CoreSwitch2	G0/0/19,20	AG-SW-2	G0/0/21,22	EtherChannel
AG-SW-1	G0/0/23,24	CoreSwitch1	G0/0/17,18	EtherChannel
AG-SW-1	G0/0/21,22	CoreSwitch2	G0/0/17,18	EtherChannel
AG-SW-2	G0/0/23,24	CoreSwitch2	G0/0/19,20	EtherChannel
AG-SW-2	G0/0/21,22	CoreSwitch1	G0/0/20,19	EtherChannel
AG-SW-2	G0/0/16	AC2	G0/0/8	
AG-SW-2	G0/0/17	File Storage 2	Ethernet0/0/0	
AG-SW-2	G0/0/18	NVR Server 2	Ethernet0/0/0	
AG-SW-2	G0/0/19	VoIP Server 2	Ethernet0/0/0	
AC2	G0/0/4	AP7	G0/0/0	
AC2	G0/0/6	AP6	G0/0/0	

ISP Chain

Device	Interface	Connected To	Interface
ISP1	G0/0/1	HQ-R1	G0/0/1
ISP1	G0/0/0	ISP2	G0/0/0
ISP2	G0/0/0	ISP1	G0/0/0
ISP2	G0/0/1	BR1-R1	G0/0/1

4. VLAN and IP Addressing Plan

VLANs include Management, IT, HR, CCTV, Guest, VOIP, Core Applications, Servers, DMZ, IoT, Backup, and WLAN SSIDs.

Each VLAN is assigned a subnet and VRRP virtual gateway on the core switches for redundancy.

DHCP pools are configured on Core-SW1 and Core-SW2 (HQ) and similarly on BR1 cores.

Here's your HQ VLAN configuration arranged in a clean, structured table:

 **HQ VLAN Configuration Table**

VLAN ID	Subnet	Default Gateway	Purpose
10	10.0.0.0/24	10.0.0.1	Network device management (switches, routers)
20	10.0.1.0/24	10.0.1.1	Network infra (NTP, DNS internal, AD DS replicas)
30	10.0.2.0/23	10.0.2.1	Application servers, DB, internal services (HQ primary)
40	10.0.4.0/24	10.0.4.1	Public-facing services (webmail, reverse-proxy, APIs)
50	10.0.5.0/24	10.0.5.1	Core business applications (ERP, finance)
60	10.0.6.0/24	10.0.6.1	HR systems / restricted systems
70	10.0.7.0/24	10.0.7.1	IT engineers / admin workstations
80	10.0.8.0/24	10.0.8.1	IP Phones (native voice VLAN)
90	10.0.9.0/24	10.0.9.1	IP cameras / video storage
100	10.0.10.0/24	10.0.10.1	Guest WIFI / internet-only clients
110	10.0.11.0/24	10.0.11.1	Dev/QA lab / sensitive environments
120	10.0.12.0/24	10.0.12.1	AP management and controller traffic
200	10.0.20.0/24	10.0.20.1	Printers, IoT sensors, building automation
250	10.0.21.0/24	10.0.21.1	Backup network / replication

Here's the VLAN configuration for  Branch 1 arranged in a structured table:

Branch 1 VLAN Configuration Table

VLAN ID	Subnet	Default Gateway	Purpose
10	10.2.0.0/24	10.2.0.1	Network device management (switches, routers)
20	10.2.1.0/24	10.2.1.1	Network infra (NTP, DNS internal, AD DS replicas)
30	10.2.2.0/23	10.2.2.1	Application servers, DB, internal services (branch servers)
40	10.2.4.0/24	10.2.4.1	Public-facing services (branch proxy, local apps)
50	10.2.5.0/24	10.2.5.1	Core business applications (ERP access, finance)
60	10.2.6.0/24	10.2.6.1	HR systems / restricted systems
70	10.2.7.0/24	10.2.7.1	IT engineers / admin workstations
80	10.2.8.0/24	10.2.8.1	IP Phones (native voice VLAN)
90	10.2.9.0/24	10.2.9.1	IP cameras / video storage
100	10.2.10.0/24	10.2.10.1	Guest WIFI / internet-only clients
110	10.2.11.0/24	10.2.11.1	Dev/QA lab / sensitive environments
120	10.2.12.0/24	10.2.12.1	AP management and controller traffic
200	10.2.20.0/24	10.2.20.1	Printers, IoT sensors, building automation
250	10.2.21.0/24	10.2.21.1	Backup network / replication

5. Routing Design

HQ and Branch are connected via a GRE tunnel established between BR1-R1 and HQ router.

Static routing ensures default route propagation to ISP next-hop.

RIP version 2 is used internally between routers to advertise internal networks through the GRE tunnel.

Routing table convergence allows full reachability between HQ subnets and BR1 subnets.

6. Redundancy & High Availability

Dual Core switches at HQ and BR1 use VRRP for gateway redundancy.

LACP EtherChannels provide link-level redundancy between core, aggregation, and access layers.

RSTP ensures loop-free topology with fast convergence.

Wireless APs are dual-registered across AC-HQ and AC-Suez for failover.

7. Wireless Network Design

Wireless network uses Huawei ACs for centralized management.

SSID 'Corelink Technologies' is broadcast using WPA2-PSK security.

APs are grouped under CORELINK-APs group with both radios broadcasting same SSID.

CAPWAP tunnels connect APs to AC-HQ or AC-Suez depending on site.

8. Security Architecture

AAA used for admin authentication on network devices.

SSH secure algorithms set (AES, SHA256).

Illegal MAC address alarms enabled across all switches.

STP edge-port disabled on trunks to prevent rogue STP influence.

DHCP exclusion ranges used to protect infrastructure addresses.

9. Testing & Validation Procedures

Connectivity Tests:

- Ping between HQ and Branch via GRE tunnel.

```
<HQ-R1>ping 10.2.0.10
PING 10.2.0.10: 56 data bytes, press CTRL_C to break
  Reply from 10.2.0.10: bytes=56 Sequence=1 ttl=255 time=20 ms
  Reply from 10.2.0.10: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.2.0.10: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 10.2.0.10: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.2.0.10: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.2.0.10 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/28/30 ms
```

```
<BR1-R1>[]
<BR1-R1>ping 10.0.0.10
PING 10.0.0.10: 56 data bytes, press CTRL_C to break
  Reply from 10.0.0.10: bytes=56 Sequence=1 ttl=255 time=30 ms
  Reply from 10.0.0.10: bytes=56 Sequence=2 ttl=255 time=30 ms
  Reply from 10.0.0.10: bytes=56 Sequence=3 ttl=255 time=40 ms
  Reply from 10.0.0.10: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.0.0.10: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.0.10 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/32/40 ms
```



```
<BR1-R1>tracert 10.0.0.10  
  
  traceroute to 10.0.0.10(10.0.0.10), max hops: 30 ,packet length: 40,press CTRL  
_C to break  
  
 1 172.16.100.1 70 ms  20 ms  30 ms
```

- Ping between VLANs inside HQ and BR1.

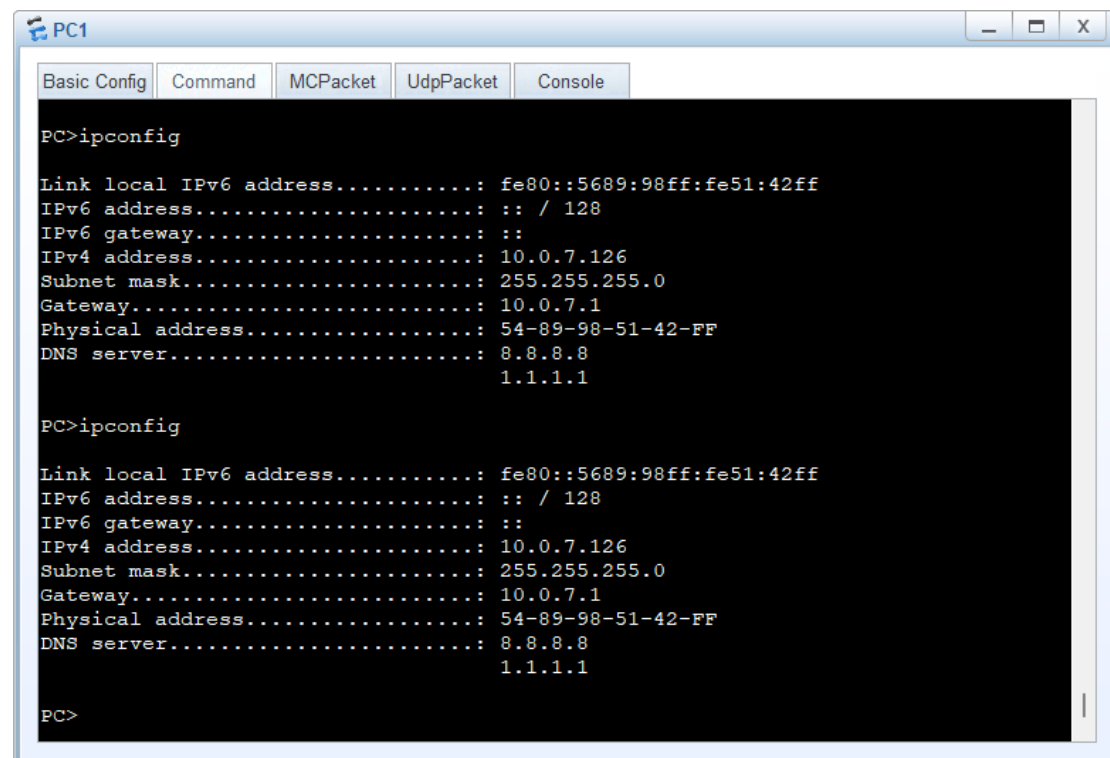
Pc 1 from HQ to FTP server in BR1

```
PC>ping 10.2.1.252

Ping 10.2.1.252: 32 data bytes, Press Ctrl_C to break
From 10.2.1.252: bytes=32 seq=1 ttl=251 time=187 ms
From 10.2.1.252: bytes=32 seq=2 ttl=251 time=157 ms
From 10.2.1.252: bytes=32 seq=3 ttl=251 time=172 ms
From 10.2.1.252: bytes=32 seq=4 ttl=251 time=141 ms
From 10.2.1.252: bytes=32 seq=5 ttl=251 time=172 ms

--- 10.2.1.252 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 141/165/187 ms
```

- DHCP address assignment validation in all VLANs.



PC1

Basic Config Command MCPacket UdpPacket Console

```
PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fe51:42ff
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.0.7.126
Subnet mask.....: 255.255.255.0
Gateway.....: 10.0.7.1
Physical address.....: 54-89-98-51-42-FF
DNS server.....: 8.8.8.8
                  1.1.1.1

PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fe51:42ff
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.0.7.126
Subnet mask.....: 255.255.255.0
Gateway.....: 10.0.7.1
Physical address.....: 54-89-98-51-42-FF
DNS server.....: 8.8.8.8
                  1.1.1.1

PC>
```

Failover Tests:

- VRRP Master/Backup role switching.

```
<Core-SW1>display vrrp brief
VRID  State      Interface      Type      Virtual IP
-----
10    Master      Vlanif10       Normal    10.0.0.1
20    Master      Vlanif20       Normal    10.0.1.1
30    Master      Vlanif30       Normal    10.0.2.1
40    Master      Vlanif40       Normal    10.0.4.1
50    Master      Vlanif50       Normal    10.0.5.1
60    Master      Vlanif60       Normal    10.0.6.1
70    Master      Vlanif70       Normal    10.0.7.1
80    Master      Vlanif80       Normal    10.0.8.1
90    Master      Vlanif90       Normal    10.0.9.1
100   Master      Vlanif100      Normal    10.0.10.1
110   Master      Vlanif110      Normal    10.0.11.1
120   Master      Vlanif120      Normal    10.0.12.1
200   Master      Vlanif200      Normal    10.0.20.1
250   Master      Vlanif250      Normal    10.0.21.1
-----
Total:14   Master:14   Backup:0      Non-active:0
<Core-SW1>
```

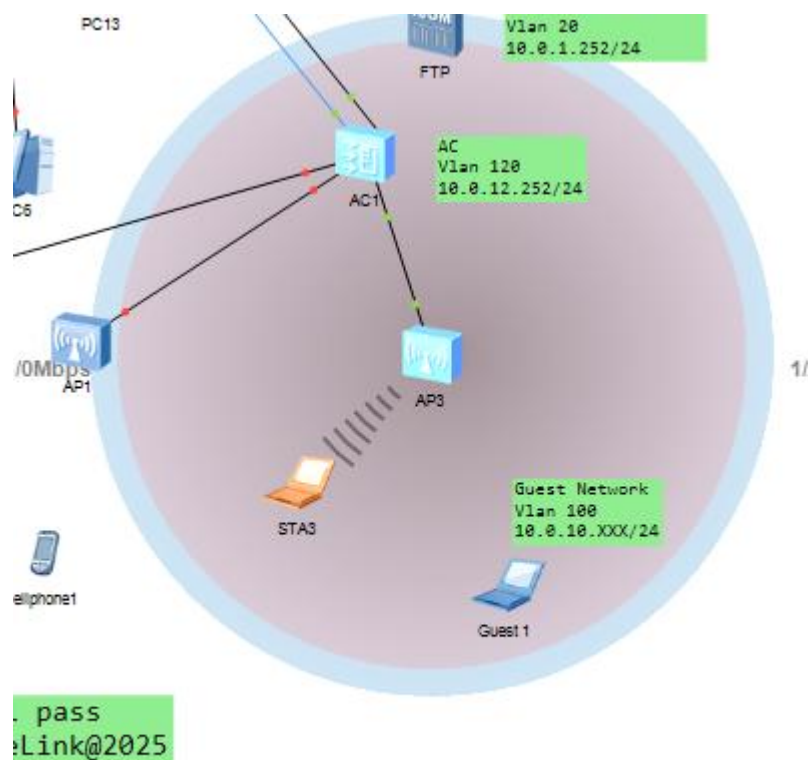
```
-----
Total:14   Master:0     Backup:14     Non-active:0
<Core-SW2>dis vrrp br
VRID  State      Interface      Type      Virtual IP
-----
10    Backup     Vlanif10       Normal    10.0.0.1
20    Backup     Vlanif20       Normal    10.0.1.1
30    Backup     Vlanif30       Normal    10.0.2.1
40    Backup     Vlanif40       Normal    10.0.4.1
50    Backup     Vlanif50       Normal    10.0.5.1
60    Backup     Vlanif60       Normal    10.0.6.1
70    Backup     Vlanif70       Normal    10.0.7.1
80    Backup     Vlanif80       Normal    10.0.8.1
90    Backup     Vlanif90       Normal    10.0.9.1
100   Backup     Vlanif100      Normal    10.0.10.1
110   Backup     Vlanif110      Normal    10.0.11.1
120   Backup     Vlanif120      Normal    10.0.12.1
200   Backup     Vlanif200      Normal    10.0.20.1
250   Backup     Vlanif250      Normal    10.0.21.1
-----
Total:14   Master:0     Backup:14     Non-active:0
<Core-SW2>
```

Wireless Tests:

- AP registration on AC.

```
<AC-HQ>dis ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor : normal      [3]
-----
ID   MAC           Name           Group          IP           Type          State
STA Uptime
-----
1    00e0-fc69-4940 Corelink-AP1   CORELINK-APs   10.0.12.151   AP2050DN      nor
0    16S
2    00e0-fcd4-63e0 Corelink-AP2   CORELINK-APs   10.0.12.209   AP2050DN      nor
0    21S
3    00e0-fcba-76d0 Corelink-AP3   CORELINK-APs   10.0.12.139   AP2050DN      nor
1    4M:15S
-----
Total: 3
```

- SSID visibility and client connectivity.



```

Welcome to use STA Simulator!

STA>ipconfig

Link local IPv6 address.....: ::
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.0.10.250
Subnet mask.....: 255.255.255.0
Gateway.....: 10.0.10.1
Physical address.....: 54-89-98-7D-19-AD
DNS server.....: 8.8.8.8
                  1.1.1.1
  
```

10. Troubleshooting Guide

Common Issues and Fixes:

- GRE Tunnel Down: Check source/destination public IPs and route to remote endpoint.
- VRRP Not Master: Verify priority settings and interface status.
- LACP Mismatch: Ensure trunk ports have identical configuration on both ends.
- DHCP Not Assigning: Confirm interface VLANIF DHCP select global/relay.
- AP Not Joining: Check CAPWAP source IP and routing to AC.

11. Maintenance Procedures

- Regular backup of switch and router configurations.
- Monitor LLDP/NDP/NTDP for topology changes.
- Verify VRRP state regularly for high availability.
- Apply firmware updates during maintenance windows.

12. Detailed Configuration Documentation

1. HQ main router

```

sysname HQ-R1
ftp server enable
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
clock timezone China-Standard-Time minus 08:00:00
portal local-server load flash:/portalpage.zip
drop illegal-mac alarmundo info-center enable
lldp enable
wlan ac-global carrier id other ac id 0
set cpu-usage threshold 80 restore 75
acl number 3000
rule 5 permit ip source 10.0.10.0 0.0.0.255
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user depi password cipher %%%$j,`W<&8x=V}=<4Lz4{/8X_S%$$$
local-user depi privilege level 10
local-user depi service-type telnet ssh ftp
local-user admin password cipher %%%$PO%0CTa%]-BUo8)@.rq%8X_1%$$$
local-user admin privilege level 15
local-user admin service-type telnet ssh ftp
local-user viewer password cipher %%%$qgRW:E5B<blJQ!fMNJ58X_L%$$$
local-user viewer privilege level 0
local-user viewer service-type telnet ssh ftp
firewall zone Local
priority 15
interface GigabitEthernet0/0/0
ip address 10.0.0.10 255.255.255.0
interface GigabitEthernet0/0/1
ip address 1.1.1.2 255.255.255.0
nat outbound 3000
interface GigabitEthernet0/0/2
interface NULL0
  
```

```
interface Tunnel0/0/0
ip address 172.16.100.1 255.255.255.252
tunnel-protocol gre
source GigabitEthernet0/0/1
destination 1.2.1.2
stelnet server enable
ip route-static 0.0.0.0 0.0.0.0 1.1.1.1
ip route-static 1.2.1.2 255.255.255.255 1.1.1.1
ip route-static 10.2.0.0 255.255.0.0 172.16.100.2
user-interface con 0
authentication-mode password
user-interface vty 0 4
authentication-mode aaa
idle-timeout 10 0
user-interface vty 16 20
wlan ac
return
<HQ-R1>
```

2. HQ Core-sw-1

```
sysname Core-SW1

FTP server enable

undo info-center enable

vlan batch 10 20 30 40 50 60 70 80 90 100

vlan batch 110 120 200 250

stp mode rstp

stp instance 0 priority 0

cluster enable

ntdp enable

ndp enable

lldp enable

undo nap slave enable

drop illegal-mac alarm

dhcp enable

diffserv domain default
```

acl number 3999

```
rule 5 permit ip source 0.0.0.0 0 destination 255.255.255.255 0
rule 10 permit ip source 10.0.0.0 0.0.0.255 destination 10.0.0.0 0.0.0.255
rule 15 permit ip source 10.0.1.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 20 permit ip source 10.0.2.0 0.0.1.255 destination 10.0.2.0 0.0.1.255
rule 25 permit ip source 10.0.4.0 0.0.0.255 destination 10.0.4.0 0.0.0.255
rule 30 permit ip source 10.0.5.0 0.0.0.255 destination 10.0.5.0 0.0.0.255
rule 35 permit ip source 10.0.6.0 0.0.0.255 destination 10.0.6.0 0.0.0.255
rule 40 permit ip source 10.0.7.0 0.0.0.255 destination 10.0.7.0 0.0.0.255
rule 45 permit ip source 10.0.8.0 0.0.0.255 destination 10.0.8.0 0.0.0.255
rule 50 permit ip source 10.0.9.0 0.0.0.255 destination 10.0.9.0 0.0.0.255
rule 55 permit ip source 10.0.10.0 0.0.0.255 destination 10.0.10.0 0.0.0.255
rule 60 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.11.0 0.0.0.255
rule 65 permit ip source 10.0.12.0 0.0.0.255 destination 10.0.12.0 0.0.0.255
rule 70 permit ip source 10.0.20.0 0.0.0.255 destination 10.0.20.0 0.0.0.255
rule 75 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.21.0 0.0.0.255
rule 80 permit ip source 10.0.0.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 85 permit ip source 10.0.0.0 0.0.0.255 destination 10.0.2.0 0.0.1.255
rule 90 permit ip source 10.0.0.0 0.0.0.255 destination 10.0.21.0 0.0.0.255
rule 95 permit ip source 10.0.2.0 0.0.1.255 destination 10.0.1.0 0.0.0.255
rule 100 permit ip source 10.0.2.0 0.0.1.255 destination 10.0.5.0 0.0.0.255
rule 105 permit ip source 10.0.2.0 0.0.1.255 destination 10.0.21.0 0.0.0.255
rule 110 permit ip source 10.0.4.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 115 permit ip source 10.0.4.0 0.0.0.255 destination 10.0.2.0 0.0.1.255
rule 120 permit ip source 10.0.5.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 125 permit ip source 10.0.5.0 0.0.0.255 destination 10.0.2.0 0.0.1.255
rule 130 permit ip source 10.0.5.0 0.0.0.255 destination 10.0.6.0 0.0.0.255
rule 135 permit ip source 10.0.6.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
```


rule 140 permit ip source 10.0.6.0 0.0.0.255 destination 10.0.5.0 0.0.0.255

rule 145 permit ip source 10.0.7.0 0.0.0.255 destination 10.0.0.0 0.0.15.255

rule 150 deny ip source 10.0.7.0 0.0.0.255 destination 10.0.10.0 0.0.0.255

rule 155 deny ip source 10.0.7.0 0.0.0.255 destination 10.0.20.0 0.0.0.255

rule 160 deny ip source 10.0.7.0 0.0.0.255 destination 10.0.21.0 0.0.0.255

rule 165 permit ip source 10.0.8.0 0.0.0.255 destination 10.0.12.0 0.0.0.255

rule 170 permit ip source 10.0.9.0 0.0.0.255 destination 10.0.21.0 0.0.0.255

rule 175 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.1.0 0.0.0.255

rule 180 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.2.0 0.0.1.255

rule 185 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.5.0 0.0.0.255

rule 190 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.21.0 0.0.0.255

rule 195 permit ip source 10.0.12.0 0.0.0.255 destination 10.0.1.0 0.0.0.255

rule 200 permit ip source 10.0.20.0 0.0.0.255 destination 10.0.1.0 0.0.0.255

rule 205 permit ip source 10.0.20.0 0.0.0.255 destination 10.0.21.0 0.0.0.255

rule 210 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.2.0 0.0.1.255

rule 215 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.5.0 0.0.0.255

rule 220 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.6.0 0.0.0.255

rule 225 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.9.0 0.0.0.255

rule 230 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.11.0 0.0.0.255

rule 235 deny ip source 10.0.10.0 0.0.0.255 destination 10.0.0.0 0.0.255.255

rule 240 deny ip source 10.0.20.0 0.0.0.255 destination 10.0.0.0 0.0.255.255

rule 245 permit ip source 10.0.1.0 0.0.0.255 destination 10.2.1.0 0.0.0.255

rule 250 permit ip source 10.0.2.0 0.0.1.255 destination 10.2.2.0 0.0.1.255

rule 255 permit ip source 10.0.5.0 0.0.0.255 destination 10.2.6.0 0.0.0.255

rule 260 permit ip source 10.0.11.0 0.0.0.255 destination 10.2.11.0 0.0.0.255

rule 265 permit ip source 10.0.21.0 0.0.0.255 destination 10.2.21.0 0.0.0.255

rule 270 deny ip

drop-profile default

```
ip pool vlan100-sw1
gateway-list 10.0.10.1
network 10.0.10.0 mask 255.255.255.0
excluded-ip-address 10.0.10.127 10.0.10.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan100-sw1
gateway-list 10.0.10.1
network 10.0.10.0 mask 255.255.255.0
excluded-ip-address 10.0.10.127 10.0.10.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan110-sw1
gateway-list 10.0.11.1
network 10.0.11.0 mask 255.255.255.0
excluded-ip-address 10.0.11.127 10.0.11.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan200-sw1
gateway-list 10.0.20.1
network 10.0.20.0 mask 255.255.255.0
excluded-ip-address 10.0.20.127 10.0.20.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan50-sw1
gateway-list 10.0.5.1
network 10.0.5.0 mask 255.255.255.0
excluded-ip-address 10.0.5.127 10.0.5.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan60-sw1
gateway-list 10.0.6.1
network 10.0.6.0 mask 255.255.255.0
```

```
excluded-ip-address 10.0.6.127 10.0.6.254

dns-list 8.8.8.8 1.1.1.1

ip pool vlan70-sw1

gateway-list 10.0.7.1

network 10.0.7.0 mask 255.255.255.0

excluded-ip-address 10.0.7.127 10.0.7.254

dns-list 8.8.8.8 1.1.1.1

ip pool vlan80-sw1

gateway-list 10.0.8.1

network 10.0.8.0 mask 255.255.255.0

excluded-ip-address 10.0.8.127 10.0.8.254

dns-list 8.8.8.8 1.1.1.1

ip pool vlan90-sw1

gateway-list 10.0.9.1

network 10.0.9.0 mask 255.255.255.0

excluded-ip-address 10.0.9.127 10.0.9.254

dns-list 8.8.8.8 1.1.1.1

aaa

authentication-scheme default

authorization-scheme default

accounting-scheme default

domain default

domain default_admin

local-user depi password cipher T70"M%@"'AC=K6%QE'XW;1!!

local-user depi privilege level 10

local-user depi service-type telnet ssh ftp

local-user admin password cipher !<*PMY9[$U)NZPO3JBXBHA!!

local-user admin privilege level 15
```

```
local-user admin service-type telnet ssh ftp
local-user viewer password cipher O.]B_=PL+/J*EP&JX-W=1!!
local-user viewer privilege level 0
local-user viewer service-type telnet ssh ftp
interface Vlanif1
interface Vlanif10
description MGMT
ip address 10.0.0.2 255.255.255.0
vrrp vrid 10 virtual-ip 10.0.0.1
vrrp vrid 10 priority 120
dhcp select global
interface Vlanif20
description INFRA
ip address 10.0.1.2 255.255.255.0
vrrp vrid 20 virtual-ip 10.0.1.1
vrrp vrid 20 priority 120
dhcp select global
interface Vlanif30
description SERVERS
ip address 10.0.2.2 255.255.254.0
vrrp vrid 30 virtual-ip 10.0.2.1
vrrp vrid 30 priority 120
dhcp select global
interface Vlanif40
description DMZ
ip address 10.0.4.2 255.255.255.0
vrrp vrid 40 virtual-ip 10.0.4.1
vrrp vrid 40 priority 120
```

```
dhcp select global

interface Vlanif50

description CORE_APPS

ip address 10.0.5.2 255.255.255.0

vrrp vrid 50 virtual-ip 10.0.5.1

vrrp vrid 50 priority 120

dhcp select global

interface Vlanif60

description HR

ip address 10.0.6.2 255.255.255.0

vrrp vrid 60 virtual-ip 10.0.6.1

vrrp vrid 60 priority 120

dhcp select global

interface Vlanif70

description IT

ip address 10.0.7.2 255.255.255.0

vrrp vrid 70 virtual-ip 10.0.7.1

vrrp vrid 70 priority 120

dhcp select global

dhcp select relay

interface Vlanif80

description VOICE

ip address 10.0.8.2 255.255.255.0

vrrp vrid 80 virtual-ip 10.0.8.1

vrrp vrid 80 priority 120

dhcp select global

interface Vlanif90

description CCTV
```

```
ip address 10.0.9.2 255.255.255.0

vrrp vrid 90 virtual-ip 10.0.9.1

vrrp vrid 90 priority 120

dhcp select global

interface Vlanif100

description GUEST

ip address 10.0.10.2 255.255.255.0

vrrp vrid 100 virtual-ip 10.0.10.1

vrrp vrid 100 priority 120

dhcp select global

interface Vlanif110

description ENGINEERS

ip address 10.0.11.2 255.255.255.0

vrrp vrid 110 virtual-ip 10.0.11.1

vrrp vrid 110 priority 120

dhcp select global

interface Vlanif120

ip address 10.0.12.2 255.255.255.0

vrrp vrid 120 virtual-ip 10.0.12.1

interface Vlanif200

description IOT

ip address 10.0.20.2 255.255.255.0

vrrp vrid 200 virtual-ip 10.0.20.1

vrrp vrid 200 priority 120

dhcp select global

interface Vlanif250

description BACKUP

ip address 10.0.21.2 255.255.255.0
```

```
vrrp vrid 250 virtual-ip 10.0.21.1  
vrrp vrid 250 priority 120  
interface MEth0/0/1  
interface Eth-Trunk1  
description Uplink-to-Core-SW2  
port link-type trunk  
port trunk allow-pass vlan 2 to 4094  
traffic-filter inbound acl 3999  
interface Eth-Trunk2  
description Agg-sw3-UPLINK  
port link-type trunk  
port trunk allow-pass vlan 2 to 4094  
traffic-filter inbound acl 3999  
interface Eth-Trunk3  
description Agg-sw1-UPLINK  
port link-type trunk  
port trunk allow-pass vlan 2 to 4094  
traffic-filter inbound acl 3999  
interface Eth-Trunk4  
traffic-filter inbound acl 3999  
interface GigabitEthernet0/0/1  
interface GigabitEthernet0/0/2  
interface GigabitEthernet0/0/3  
interface GigabitEthernet0/0/4  
interface GigabitEthernet0/0/5  
interface GigabitEthernet0/0/6  
interface GigabitEthernet0/0/7  
interface GigabitEthernet0/0/8
```

```
interface GigabitEthernet0/0/9

interface GigabitEthernet0/0/10

interface GigabitEthernet0/0/11

interface GigabitEthernet0/0/12

interface GigabitEthernet0/0/13

interface GigabitEthernet0/0/14

interface GigabitEthernet0/0/15

eth-trunk 2

interface GigabitEthernet0/0/16

eth-trunk 2

interface GigabitEthernet0/0/17

interface GigabitEthernet0/0/18

interface GigabitEthernet0/0/19

eth-trunk 3

interface GigabitEthernet0/0/20

eth-trunk 3

interface GigabitEthernet0/0/21

interface GigabitEthernet0/0/22

port link-type access

port default vlan 10

interface GigabitEthernet0/0/23

eth-trunk 1

interface GigabitEthernet0/0/24

eth-trunk 1

interface NULL0

ip route-static 0.0.0.0 0.0.0.0 10.0.0.10

ip route-static 10.2.0.0 255.255.0.0 172.16.100.1

ip route-static 172.16.100.0 255.255.255.252 10.0.0.10
```



```
stelnet server enable  
user-interface con 0  
user-interface vty 0 4  
authentication-mode aaa  
return  
<Core-SW1>
```

3. HQ Core-Sw-2

```
sysname Core-SW2  
FTP server enable  
undo info-center enable  
vlan batch 10 20 30 40 50 60 70 80 90 100  
vlan batch 110 120 200 250  
stp mode rstp  
stp instance 0 priority 4096  
cluster enable  
ntdp enable  
ndp enable  
lldp enable  
undo nap slave enable  
drop illegal-mac alarm  
dhcp enable  
diffserv domain default  
acl number 3999  
rule 5 permit ip source 0.0.0.0 0 destination 255.255.255.255 0  
rule 10 permit ip source 10.0.0.0 0.0.0.255 destination 10.0.0.0 0.0.0.255  
rule 15 permit ip source 10.0.1.0 0.0.0.255 destination 10.0.1.0 0.0.0.255  
rule 20 permit ip source 10.0.2.0 0.0.1.255 destination 10.0.2.0 0.0.1.255  
rule 25 permit ip source 10.0.4.0 0.0.0.255 destination 10.0.4.0 0.0.0.255
```

rule 30 permit ip source 10.0.5.0 0.0.0.255 destination 10.0.5.0 0.0.0.255

rule 35 permit ip source 10.0.6.0 0.0.0.255 destination 10.0.6.0 0.0.0.255

rule 40 permit ip source 10.0.7.0 0.0.0.255 destination 10.0.7.0 0.0.0.255

rule 45 permit ip source 10.0.8.0 0.0.0.255 destination 10.0.8.0 0.0.0.255

rule 50 permit ip source 10.0.9.0 0.0.0.255 destination 10.0.9.0 0.0.0.255

rule 55 permit ip source 10.0.10.0 0.0.0.255 destination 10.0.10.0 0.0.0.255

rule 60 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.11.0 0.0.0.255

rule 65 permit ip source 10.0.12.0 0.0.0.255 destination 10.0.12.0 0.0.0.255

rule 70 permit ip source 10.0.20.0 0.0.0.255 destination 10.0.20.0 0.0.0.255

rule 75 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.21.0 0.0.0.255

rule 80 permit ip source 10.0.0.0 0.0.0.255 destination 10.0.1.0 0.0.0.255

rule 85 permit ip source 10.0.0.0 0.0.0.255 destination 10.0.2.0 0.0.1.255

rule 90 permit ip source 10.0.0.0 0.0.0.255 destination 10.0.21.0 0.0.0.255

rule 95 permit ip source 10.0.2.0 0.0.1.255 destination 10.0.1.0 0.0.0.255

rule 100 permit ip source 10.0.2.0 0.0.1.255 destination 10.0.5.0 0.0.0.255

rule 105 permit ip source 10.0.2.0 0.0.1.255 destination 10.0.21.0 0.0.0.255

rule 110 permit ip source 10.0.4.0 0.0.0.255 destination 10.0.1.0 0.0.0.255

rule 115 permit ip source 10.0.4.0 0.0.0.255 destination 10.0.2.0 0.0.1.255

rule 120 permit ip source 10.0.5.0 0.0.0.255 destination 10.0.1.0 0.0.0.255

rule 125 permit ip source 10.0.5.0 0.0.0.255 destination 10.0.2.0 0.0.1.255

rule 130 permit ip source 10.0.5.0 0.0.0.255 destination 10.0.6.0 0.0.0.255

rule 135 permit ip source 10.0.6.0 0.0.0.255 destination 10.0.1.0 0.0.0.255

rule 140 permit ip source 10.0.6.0 0.0.0.255 destination 10.0.5.0 0.0.0.255

rule 145 permit ip source 10.0.7.0 0.0.0.255 destination 10.0.0.0 0.0.15.255

rule 150 deny ip source 10.0.7.0 0.0.0.255 destination 10.0.10.0 0.0.0.255

rule 155 deny ip source 10.0.7.0 0.0.0.255 destination 10.0.20.0 0.0.0.255

rule 160 deny ip source 10.0.7.0 0.0.0.255 destination 10.0.21.0 0.0.0.255

rule 165 permit ip source 10.0.8.0 0.0.0.255 destination 10.0.12.0 0.0.0.255

```
rule 170 permit ip source 10.0.9.0 0.0.0.255 destination 10.0.21.0 0.0.0.255
rule 175 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 180 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.2.0 0.0.1.255
rule 185 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.5.0 0.0.0.255
rule 190 permit ip source 10.0.11.0 0.0.0.255 destination 10.0.21.0 0.0.0.255
rule 195 permit ip source 10.0.12.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 200 permit ip source 10.0.20.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 205 permit ip source 10.0.20.0 0.0.0.255 destination 10.0.21.0 0.0.0.255
rule 210 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.2.0 0.0.1.255
rule 215 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.5.0 0.0.0.255
rule 220 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.6.0 0.0.0.255
rule 225 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.9.0 0.0.0.255
rule 230 permit ip source 10.0.21.0 0.0.0.255 destination 10.0.11.0 0.0.0.255
rule 235 deny ip source 10.0.10.0 0.0.0.255 destination 10.0.0.0 0.0.255.255
rule 240 deny ip source 10.0.20.0 0.0.0.255 destination 10.0.0.0 0.0.255.255
rule 245 permit ip source 10.0.1.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 250 permit ip source 10.0.2.0 0.0.1.255 destination 10.2.2.0 0.0.1.255
rule 255 permit ip source 10.0.5.0 0.0.0.255 destination 10.2.6.0 0.0.0.255
rule 260 permit ip source 10.0.11.0 0.0.0.255 destination 10.2.11.0 0.0.0.255
rule 265 permit ip source 10.0.21.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 270 deny ip
drop-profile default
ip pool vlan100-sw2
gateway-list 10.0.10.1
network 10.0.10.0 mask 255.255.255.0
excluded-ip-address 10.0.10.251 10.0.10.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan110-sw2
```

```
gateway-list 10.0.11.1
network 10.0.11.0 mask 255.255.255.0
excluded-ip-address 10.0.11.251 10.0.11.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan200-sw2
gateway-list 10.0.20.1
network 10.0.20.0 mask 255.255.255.0
excluded-ip-address 10.0.20.251 10.0.20.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan50-sw2
gateway-list 10.0.5.1
network 10.0.5.0 mask 255.255.255.0
excluded-ip-address 10.0.5.251 10.0.5.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan60-sw2
gateway-list 10.0.6.1
network 10.0.6.0 mask 255.255.255.0
excluded-ip-address 10.0.6.251 10.0.6.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan70-sw2
gateway-list 10.0.7.1
network 10.0.7.0 mask 255.255.255.0
excluded-ip-address 10.0.7.251 10.0.7.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan80-sw2
gateway-list 10.0.8.1
network 10.0.8.0 mask 255.255.255.0
excluded-ip-address 10.0.8.251 10.0.8.254
```

```
dns-list 8.8.8.8 1.1.1.1

ip pool vlan90-sw2

gateway-list 10.0.9.1

network 10.0.9.0 mask 255.255.255.0

excluded-ip-address 10.0.9.251 10.0.9.254

dns-list 8.8.8.8 1.1.1.1

aaa

authentication-scheme default

authorization-scheme default

accounting-scheme default

domain default

domain default_admin

local-user depi password cipher T70"M%@"'AC=K6%QE'XW;1!!

local-user depi privilege level 10

local-user depi service-type telnet ssh ftp

local-user admin password cipher !<*PMY9[$U)NZPO3J)B)BHA!!

local-user admin privilege level 15

local-user admin service-type telnet ssh ftp

local-user viewer password cipher O.]B_=PL+ /J*EP&JX-W=1!!

local-user viewer privilege level 0

local-user viewer service-type telnet ssh ftp

interface Vlanif1

interface Vlanif10

description MGMT

ip address 10.0.0.3 255.255.255.0

vrrp vrid 10 virtual-ip 10.0.0.1

interface Vlanif20

description INFRA
```

```
ip address 10.0.1.3 255.255.255.0

vrrp vrid 20 virtual-ip 10.0.1.1

interface Vlanif30

description SERVERS

ip address 10.0.2.3 255.255.254.0

vrrp vrid 30 virtual-ip 10.0.2.1

interface Vlanif40

description DMZ

ip address 10.0.4.3 255.255.255.0

vrrp vrid 40 virtual-ip 10.0.4.1

interface Vlanif50

description CORE_APPS

ip address 10.0.5.3 255.255.255.0

vrrp vrid 50 virtual-ip 10.0.5.1

dhcp select global

interface Vlanif60

description HR

ip address 10.0.6.3 255.255.255.0

vrrp vrid 60 virtual-ip 10.0.6.1

dhcp select global

interface Vlanif70

description IT

ip address 10.0.7.3 255.255.255.0

vrrp vrid 70 virtual-ip 10.0.7.1

dhcp select global

interface Vlanif80

description VOICE

ip address 10.0.8.3 255.255.255.0
```

```
vrrp vrid 80 virtual-ip 10.0.8.1  
dhcp select global  
interface Vlanif90  
description CCTV  
ip address 10.0.9.3 255.255.255.0  
vrrp vrid 90 virtual-ip 10.0.9.1  
dhcp select global  
interface Vlanif100  
description GUEST  
ip address 10.0.10.3 255.255.255.0  
vrrp vrid 100 virtual-ip 10.0.10.1  
dhcp select global  
interface Vlanif110  
description ENGINEERS  
ip address 10.0.11.3 255.255.255.0  
vrrp vrid 110 virtual-ip 10.0.11.1  
dhcp select global  
interface Vlanif120  
ip address 10.0.12.3 255.255.255.0  
vrrp vrid 120 virtual-ip 10.0.12.1  
interface Vlanif200  
description IOT  
ip address 10.0.20.3 255.255.255.0  
vrrp vrid 200 virtual-ip 10.0.20.1  
dhcp select global  
interface Vlanif250  
ip address 10.0.21.3 255.255.255.0  
vrrp vrid 250 virtual-ip 10.0.21.1
```

```
interface MEth0/0/1

interface Eth-Trunk1

description Uplink-to-Core-SW1

port link-type trunk

port trunk allow-pass vlan 2 to 4094

traffic-filter inbound acl 3999

interface Eth-Trunk2

description Agg-sw3-UPLINK

port link-type trunk

port trunk allow-pass vlan 2 to 4094

traffic-filter inbound acl 3999

interface Eth-Trunk3

description Agg-sw1-UPLINK

port link-type trunk

port trunk allow-pass vlan 2 to 4094

traffic-filter inbound acl 3999

interface GigabitEthernet0/0/1

interface GigabitEthernet0/0/2

interface GigabitEthernet0/0/3

interface GigabitEthernet0/0/4

interface GigabitEthernet0/0/5

interface GigabitEthernet0/0/6

interface GigabitEthernet0/0/7

interface GigabitEthernet0/0/8

interface GigabitEthernet0/0/9

interface GigabitEthernet0/0/10

interface GigabitEthernet0/0/11

interface GigabitEthernet0/0/12
```



```
interface GigabitEthernet0/0/13

interface GigabitEthernet0/0/14

interface GigabitEthernet0/0/15

eth-trunk 3

interface GigabitEthernet0/0/16

eth-trunk 3

interface GigabitEthernet0/0/17

interface GigabitEthernet0/0/18

interface GigabitEthernet0/0/19

eth-trunk 2

interface GigabitEthernet0/0/20

eth-trunk 2

interface GigabitEthernet0/0/21

interface GigabitEthernet0/0/22

port link-type access

port default vlan 10

interface GigabitEthernet0/0/23

eth-trunk 1

interface GigabitEthernet0/0/24

eth-trunk 1

interface NULL0

ip route-static 0.0.0.0 0.0.0.0 10.0.0.10

ip route-static 10.2.0.0 255.255.0.0 10.0.0.10

ip route-static 10.2.0.0 255.255.0.0 172.16.100.1

ip route-static 172.16.100.0 255.255.255.252 10.0.0.10

stelnet server enable

user-interface con 0

user-interface vty 0 4
```

authentication-mode aaa

return

4. HQ Agg-sw-1

```
sysname Agg-sw1
undo info-center enable
vlan batch 10 20 30 40 50 60 70 80 90 100
vlan batch 110 120 200 250
stp mode rstp
cluster enable
ntdp enable
ndp enable
drop illegal-mac alarm
dhcp enable
dhcp snooping enable
diffserv domain default
drop-profile default
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password simple admin
local-user admin service-type http
interface Vlanif1
interface MEth0/0/1
interface Eth-Trunk1
description Core-sw-1-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
stp edged-port disable
dhcp snooping trusted
interface Eth-Trunk2
description Core-sw-2-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
stp edged-port disable
dhcp snooping trusted
interface GigabitEthernet0/0/1
port link-type access
port default vlan 70
interface GigabitEthernet0/0/2
port link-type access
port default vlan 60
interface GigabitEthernet0/0/3
```

```
port link-type access
port default vlan 200
interface GigabitEthernet0/0/4
interface GigabitEthernet0/0/5
interface GigabitEthernet0/0/6
interface GigabitEthernet0/0/7
interface GigabitEthernet0/0/8
interface GigabitEthernet0/0/9
interface GigabitEthernet0/0/10
interface GigabitEthernet0/0/11
interface GigabitEthernet0/0/12
interface GigabitEthernet0/0/13
interface GigabitEthernet0/0/14
interface GigabitEthernet0/0/15
interface GigabitEthernet0/0/16
interface GigabitEthernet0/0/17
interface GigabitEthernet0/0/18
interface GigabitEthernet0/0/19
interface GigabitEthernet0/0/20
interface GigabitEthernet0/0/21
eth-trunk 2
interface GigabitEthernet0/0/22
eth-trunk 2
interface GigabitEthernet0/0/23
eth-trunk 1
interface GigabitEthernet0/0/24
eth-trunk 1
interface NULL0
user-interface con 0
user-interface vty 0 4
return
```

5. HQ-Agg-sw-2

```
sysname Agg-SW2
undo info-center enable
vlan batch 10 20 30 40 50 60 70 80 90 100
vlan batch 110 120 200 250
stp mode rstp
cluster enable
ntdp enable
ndp enable
drop illegal-mac alarm
dhcp enable
dhcp snooping enable
diffserv domain default
drop-profile default
aaa
authentication-scheme default
```

```
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password simple admin
local-user admin service-type http
interface Vlanif1
interface MEth0/0/1
interface Eth-Trunk1
description Core-sw-1-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
stp edged-port disable
dhcp snooping trusted
interface Eth-Trunk2
description Core-sw-2-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
stp edged-port disable
dhcp snooping trusted
interface GigabitEthernet0/0/1
port link-type access
port default vlan 70
interface GigabitEthernet0/0/2
port link-type access
port default vlan 110
interface GigabitEthernet0/0/3
port link-type access
port default vlan 60
interface GigabitEthernet0/0/4
interface GigabitEthernet0/0/5
interface GigabitEthernet0/0/6
interface GigabitEthernet0/0/7
interface GigabitEthernet0/0/8
interface GigabitEthernet0/0/9
interface GigabitEthernet0/0/10
interface GigabitEthernet0/0/11
interface GigabitEthernet0/0/12
interface GigabitEthernet0/0/13
interface GigabitEthernet0/0/14
interface GigabitEthernet0/0/15
interface GigabitEthernet0/0/16
interface GigabitEthernet0/0/17
interface GigabitEthernet0/0/18
interface GigabitEthernet0/0/19
interface GigabitEthernet0/0/20
interface GigabitEthernet0/0/21
```

```
eth-trunk 2
interface GigabitEthernet0/0/22
eth-trunk 2
interface GigabitEthernet0/0/23
eth-trunk 1
interface GigabitEthernet0/0/24
eth-trunk 1
interface NULL0
user-interface con 0
user-interface vty 0 4
return
```

6. HQ-Agg-sw-3

```
sysname Agg-SW3
undo info-center enable
vlan batch 10 20 30 40 50 60 70 80 90 100
vlan batch 110 120 200 250
stp mode rstp
cluster enable
ntdp enable
ndp enable
drop illegal-mac alarm
dhcp enable
dhcp snooping enable
diffserv domain default
drop-profile default
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password simple admin
local-user admin service-type http
interface Vlanif1
interface MEth0/0/1
interface Eth-Trunk1
description Core-sw1-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
stp edged-port disable
dhcp snooping trusted
interface Eth-Trunk2
description Core-sw2-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
dhcp snooping trusted
interface Eth-Trunk3
```

```
description AP-UPLINK
port link-type trunk
port trunk allow-pass vlan 100 120
interface GigabitEthernet0/0/1
description File-Storage-UPLINK
port link-type access
port default vlan 20
interface GigabitEthernet0/0/2
description NVR-UPLINK
port link-type access
port default vlan 90
interface GigabitEthernet0/0/3
description VOIP-UPLINK
port link-type access
port default vlan 80
interface GigabitEthernet0/0/4
port link-type access
port default vlan 70
interface GigabitEthernet0/0/5
interface GigabitEthernet0/0/6
interface GigabitEthernet0/0/7
interface GigabitEthernet0/0/8
interface GigabitEthernet0/0/9
interface GigabitEthernet0/0/10
interface GigabitEthernet0/0/11
interface GigabitEthernet0/0/12
interface GigabitEthernet0/0/13
interface GigabitEthernet0/0/14
interface GigabitEthernet0/0/15
interface GigabitEthernet0/0/16
interface GigabitEthernet0/0/17
interface GigabitEthernet0/0/18
description DHCP-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
interface GigabitEthernet0/0/19
eth-trunk 3
interface GigabitEthernet0/0/20
eth-trunk 3
interface GigabitEthernet0/0/21
eth-trunk 2
interface GigabitEthernet0/0/22
eth-trunk 2
interface GigabitEthernet0/0/23
eth-trunk 1
interface GigabitEthernet0/0/24
eth-trunk 1
```

```
interface NULL0
user-interface con 0
user-interface vty 0 4
return
```

7. HQ-AC-1

```
sysname AC-HQ
undo http secure-server enable
set memory-usage threshold 0
ssl renegotiation-rate 1
vlan batch 100 120
authentication-profile name default_authen_profile
authentication-profile name dot1x_authen_profile
authentication-profile name mac_authen_profile
authentication-profile name portal_authen_profile
authentication-profile name macportal_authen_profile
dhcp enable
dhcp snooping enable
diffserv domain default
radius-server template default
pki realm default
rsa local-key-pair default
enrollment self-signed
ike proposal default
encryption-algorithm aes-256
dh group14
authentication-algorithm sha2-256
authentication-method pre-share
integrity-algorithm hmac-sha2-256
prf hmac-sha2-256
free-rule-template name default_free_rule
portal-access-profile name portal_access_profile
ip pool vlan120
gateway-list 10.0.12.1
network 10.0.12.0 mask 255.255.255.0
dns-list 8.8.8.8 8.8.4.4
aaa
authentication-scheme default
authentication-scheme radius
authentication-mode radius
authorization-scheme default
accounting-scheme default
domain default
authentication-scheme radius
radius-server default
domain default_admin
authentication-scheme default
```

```
local-user admin password irreversible-cipher
$1a$7(vSY/0I*,$zcpkA=T\W6Vtum2'.y}08bDuT].C&$*;,d8@I*5S$
local-user admin service-type http
interface Vlanif120
ip address 10.0.12.252 255.255.255.0
dhcp select global
interface Eth-Trunk1
description AGG-SW3-UPLINK
port link-type trunk
port trunk allow-pass vlan 100 120
dhcp snooping trusted
interface Eth-Trunk2
description AP-1-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface Eth-Trunk3
description AP-2-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface Eth-Trunk4
description AP-3-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface GigabitEthernet0/0/1
eth-trunk 4
interface GigabitEthernet0/0/2
eth-trunk 4
interface GigabitEthernet0/0/3
eth-trunk 3
interface GigabitEthernet0/0/4
eth-trunk 3
interface GigabitEthernet0/0/5
eth-trunk 2
interface GigabitEthernet0/0/6
eth-trunk 2
interface GigabitEthernet0/0/7
undo negotiation auto
duplex half
eth-trunk 1
interface GigabitEthernet0/0/8
undo negotiation auto
duplex half
eth-trunk 1
interface NULL0
```



```

snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
ssh server secure-algorithms cipher aes256_ctr aes128_ctr
ssh server key-exchange dh_group14_sha1
ssh client secure-algorithms cipher aes256_ctr aes128_ctr
ssh client secure-algorithms hmac sha2_256
ssh client key-exchange dh_group14_sha1
ip route-static 0.0.0.0 0.0.0.0 10.0.12.1
capwap source ip-address 10.0.12.252
user-interface con 0
user-interface vty 0 4
protocol inbound all
user-interface vty 16 20
protocol inbound all
wlan
traffic-profile name default
security-profile name default
security-profile name default-wds
security-profile name Corelink-Sec
security wpa-wpa2 psk pass-phrase
%^%#;ov#(gw+_h!FeM#1SSI(#DWGrV(o1Uhxm&K!yoT%^%# aes
security-profile name default-mesh
ssid-profile name default
ssid-profile name Corelink-SSID
ssid "Corelink Technologies"
vap-profile name default
vap-profile name Corelink-VAP
service-vlan vlan-id 100
ssid-profile Corelink-SSID
security-profile Corelink-Sec
wds-profile name default
mesh-handover-profile name default
mesh-profile name default
regulatory-domain-profile name default
air-scan-profile name default
rrm-profile name default
radio-2g-profile name default
radio-5g-profile name default
wids-spoof-profile name default
wids-profile name default
wireless-access-specification
ap-system-profile name default
port-link-profile name default
wired-port-profile name default
serial-profile name preset-enjoyor-toeap
ap-group name default
ap-group name CORELINK-APs

```

```
radio 0
 vap-profile Corelink-VAP wlan 1
radio 1
 vap-profile Corelink-VAP wlan 1
ap-id 1 type-id 69 ap-mac 00e0-fc69-4940 ap-sn 2102354483102A1F6A74
ap-name Corelink-AP1
ap-group CORELINK-APs
ap-id 2 type-id 69 ap-mac 00e0-fcd4-63e0 ap-sn 2102354483105A412F21
ap-name Corelink-AP2
ap-group CORELINK-APs
ap-id 3 type-id 69 ap-mac 00e0-fcba-76d0 ap-sn 2102354483108C3A7737
ap-name Corelink-AP3
ap-group CORELINK-APs
provision-ap
dot1x-access-profile name dot1x_access_profile
mac-access-profile name mac_access_profile
return
```

8. Branch main router

```
sysname AC-HQ
undo http secure-server enable
set memory-usage threshold 0
ssl renegotiation-rate 1
vlan batch 100 120
authentication-profile name default_authen_profile
authentication-profile name dot1x_authen_profile
authentication-profile name mac_authen_profile
authentication-profile name portal_authen_profile
authentication-profile name macportal_authen_profile
dhcp enable
dhcp snooping enable
diffserv domain default
radius-server template default
pki realm default
rsa local-key-pair default
enrollment self-signed
ike proposal default
encryption-algorithm aes-256
dh group14
authentication-algorithm sha2-256
authentication-method pre-share
integrity-algorithm hmac-sha2-256
prf hmac-sha2-256
free-rule-template name default_free_rule
```

```
portal-access-profile name portal_access_profile
ip pool vlan120
gateway-list 10.0.12.1
network 10.0.12.0 mask 255.255.255.0
dns-list 8.8.8.8 8.8.4.4
aaa
authentication-scheme default
authentication-scheme radius
authentication-mode radius
authorization-scheme default
accounting-scheme default
domain default
authentication-scheme radius
radius-server default
domain default_admin
authentication-scheme default
local-user admin password irreversible-cipher
$1a$7(vSY/OI*,$zcpkA=T\W6Vtum2'.y}O8bDuT].C&$*;,d8@I*5S$
local-user admin service-type http
interface Vlanif120
ip address 10.0.12.252 255.255.255.0
dhcp select global
interface Eth-Trunk1
description AGG-SW3-UPLINK
port link-type trunk
port trunk allow-pass vlan 100 120
dhcp snooping trusted
interface Eth-Trunk2
description AP-1-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface Eth-Trunk3
description AP-2-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface Eth-Trunk4
description AP-3-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface GigabitEthernet0/0/1
eth-trunk 4
interface GigabitEthernet0/0/2
eth-trunk 4
interface GigabitEthernet0/0/3
```

```

eth-trunk 3
interface GigabitEthernet0/0/4
eth-trunk 3
interface GigabitEthernet0/0/5
eth-trunk 2
interface GigabitEthernet0/0/6
eth-trunk 2
interface GigabitEthernet0/0/7
undo negotiation auto
duplex half
eth-trunk 1
interface GigabitEthernet0/0/8
undo negotiation auto
duplex half
eth-trunk 1
interface NULL0
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
ssh server secure-algorithms cipher aes256_ctr aes128_ctr
ssh server key-exchange dh_group14_sha1
ssh client secure-algorithms cipher aes256_ctr aes128_ctr
ssh client secure-algorithms hmac sha2_256
ssh client key-exchange dh_group14_sha1
ip route-static 0.0.0.0 0.0.0.0 10.0.12.1
capwap source ip-address 10.0.12.252
user-interface con 0
user-interface vty 0 4
protocol inbound all
user-interface vty 16 20
protocol inbound all
wlan
traffic-profile name default
security-profile name default
security-profile name default-wds
security-profile name Corelink-Sec
security wpa-wpa2 psk pass-phrase
%^%#;ov#(gw+_h!FeM#1SSI(#DWGrV(o1Uhxm&K!yoT%^%# aes
security-profile name default-mesh
ssid-profile name default
ssid-profile name Corelink-SSID
ssid "Corelink Technologies"
vap-profile name default
vap-profile name Corelink-VAP
service-vlan vlan-id 100
ssid-profile Corelink-SSID
security-profile Corelink-Sec
wds-profile name default

```

```

mesh-handover-profile name default
mesh-profile name default
regulatory-domain-profile name default
air-scan-profile name default
rrm-profile name default
radio-2g-profile name default
radio-5g-profile name default
wids-spoof-profile name default
wids-profile name default
wireless-access-specification
ap-system-profile name default
port-link-profile name default
wired-port-profile name default
serial-profile name preset-enjoyor-toeap
ap-group name default
ap-group name CORELINK-APs
radio 0
  vap-profile Corelink-VAP wlan 1
radio 1
  vap-profile Corelink-VAP wlan 1
ap-id 1 type-id 69 ap-mac 00e0-fc69-4940 ap-sn 2102354483102A1F6A74
ap-name Corelink-AP1
ap-group CORELINK-APs
ap-id 2 type-id 69 ap-mac 00e0-fcd4-63e0 ap-sn 2102354483105A412F21
ap-name Corelink-AP2
ap-group CORELINK-APs
ap-id 3 type-id 69 ap-mac 00e0-fcba-76d0 ap-sn 2102354483108C3A7737
ap-name Corelink-AP3
ap-group CORELINK-APs
provision-ap
dot1x-access-profile name dot1x_access_profile
mac-access-profile name mac_access_profile
return
  
```

9. BR1-core-sw-1

```

sysname BR1-Coreswitch-2
FTP server enable
undo info-center enable
vlan batch 10 20 30 40 50 60 70 80 90 100
vlan batch 110 120 200 250
stp mode rstp
stp instance 0 priority 0
cluster enable
ntdp enable
ndp enable
undo nap slave enable
drop illegal-mac alarm
dhcp enable
  
```

```
diffserv domain default
acl number 3999
rule 5 permit ip source 0.0.0.0 0 destination 255.255.255.255 0
rule 10 permit ip source 10.2.0.0 0.0.0.255 destination 10.2.0.0 0.0.0.255
rule 15 permit ip source 10.2.1.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 20 permit ip source 10.2.2.0 0.0.1.255 destination 10.2.2.0 0.0.1.255
rule 25 permit ip source 10.2.4.0 0.0.0.255 destination 10.2.4.0 0.0.0.255
rule 30 permit ip source 10.2.5.0 0.0.0.255 destination 10.2.5.0 0.0.0.255
rule 35 permit ip source 10.2.6.0 0.0.0.255 destination 10.2.6.0 0.0.0.255
rule 40 permit ip source 10.2.7.0 0.0.0.255 destination 10.2.7.0 0.0.0.255
rule 45 permit ip source 10.2.8.0 0.0.0.255 destination 10.2.8.0 0.0.0.255
rule 50 permit ip source 10.2.9.0 0.0.0.255 destination 10.2.9.0 0.0.0.255
rule 55 permit ip source 10.2.10.0 0.0.0.255 destination 10.2.10.0 0.0.0.255
rule 60 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.11.0 0.0.0.255
rule 65 permit ip source 10.2.12.0 0.0.0.255 destination 10.2.12.0 0.0.0.255
rule 70 permit ip source 10.2.20.0 0.0.0.255 destination 10.2.20.0 0.0.0.255
rule 75 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 80 permit ip source 10.2.0.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 85 permit ip source 10.2.0.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 90 permit ip source 10.2.0.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 95 permit ip source 10.2.2.0 0.0.1.255 destination 10.2.1.0 0.0.0.255
rule 100 permit ip source 10.2.2.0 0.0.1.255 destination 10.2.5.0 0.0.0.255
rule 105 permit ip source 10.2.2.0 0.0.1.255 destination 10.2.21.0 0.0.0.255
rule 110 permit ip source 10.2.4.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 115 permit ip source 10.2.4.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 120 permit ip source 10.2.5.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 125 permit ip source 10.2.5.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 130 permit ip source 10.2.5.0 0.0.0.255 destination 10.2.6.0 0.0.0.255
rule 135 permit ip source 10.2.6.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 140 permit ip source 10.2.6.0 0.0.0.255 destination 10.2.5.0 0.0.0.255
rule 145 permit ip source 10.2.7.0 0.0.0.255 destination 10.2.0.0 0.0.31.255
rule 150 deny ip source 10.2.7.0 0.0.0.255 destination 10.2.10.0 0.0.0.255
rule 155 deny ip source 10.2.7.0 0.0.0.255 destination 10.2.20.0 0.0.0.255
rule 160 deny ip source 10.2.7.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 165 permit ip source 10.2.8.0 0.0.0.255 destination 10.2.12.0 0.0.0.255
rule 170 permit ip source 10.2.9.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 175 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 180 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 185 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.5.0 0.0.0.255
rule 190 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 195 permit ip source 10.2.12.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 200 permit ip source 10.2.20.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 205 permit ip source 10.2.20.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 210 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 215 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.5.0 0.0.0.255
rule 220 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.6.0 0.0.0.255
rule 225 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.9.0 0.0.0.255
```

```
rule 230 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.11.0 0.0.0.255
rule 235 deny ip source 10.2.10.0 0.0.0.255 destination 10.2.0.0 0.0.255.255
rule 240 deny ip source 10.2.20.0 0.0.0.255 destination 10.2.0.0 0.0.255.255
rule 245 deny ip source 10.2.11.0 0.0.0.255 destination 10.2.10.0 0.0.0.255
rule 250 permit ip source 10.2.1.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 255 permit ip source 10.2.2.0 0.0.1.255 destination 10.0.2.0 0.0.1.255
rule 260 permit ip source 10.2.6.0 0.0.0.255 destination 10.0.5.0 0.0.0.255
rule 265 permit ip source 10.2.11.0 0.0.0.255 destination 10.0.11.0 0.0.0.255
rule 270 permit ip source 10.2.21.0 0.0.0.255 destination 10.0.21.0 0.0.0.255
rule 275 deny ip
drop-profile default
ip pool vlan100-sw2
gateway-list 10.2.10.1
network 10.2.10.0 mask 255.255.255.0
excluded-ip-address 10.2.10.251 10.2.10.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan110-sw2
gateway-list 10.2.11.1
network 10.2.11.0 mask 255.255.255.0
excluded-ip-address 10.2.11.251 10.2.11.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan200-sw2
gateway-list 10.2.20.1
network 10.2.20.0 mask 255.255.255.0
excluded-ip-address 10.2.20.251 10.2.20.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan50-sw2
gateway-list 10.2.5.1
network 10.2.5.0 mask 255.255.255.0
excluded-ip-address 10.2.5.251 10.2.5.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan60-sw2
gateway-list 10.2.6.1
network 10.2.6.0 mask 255.255.255.0
excluded-ip-address 10.2.6.251 10.2.6.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan70-sw2
gateway-list 10.2.7.1
network 10.2.7.0 mask 255.255.255.0
excluded-ip-address 10.2.7.251 10.2.7.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan80-sw2
gateway-list 10.2.8.1
network 10.2.8.0 mask 255.255.255.0
excluded-ip-address 10.2.8.251 10.2.8.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan90-sw2
```

```
gateway-list 10.2.9.1
network 10.2.9.0 mask 255.255.255.0
excluded-ip-address 10.2.9.251 10.2.9.254
dns-list 8.8.8.8 1.1.1.1
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user depi password cipher T70"M%@"'AC=K6%QE'XW;1!!
local-user depi privilege level 10
local-user depi service-type telnet ssh ftp
local-user admin password cipher !<*PMY9[$U)NZPO3JBXBHA!!
local-user admin privilege level 15
local-user admin service-type telnet ssh ftp
local-user viewer password cipher O.]B_=PL+/J*EP&JX-W=1!!
local-user viewer privilege level 0
local-user viewer service-type telnet ssh ftp
interface Vlanif1
interface Vlanif10
description MGMT
ip address 10.2.0.3 255.255.255.0
vrrp vrid 10 virtual-ip 10.2.0.1
dhcp select global
interface Vlanif20
description INFRA
ip address 10.2.1.3 255.255.255.0
vrrp vrid 20 virtual-ip 10.2.1.1
dhcp select global
interface Vlanif30
description SERVERS
ip address 10.2.2.3 255.255.254.0
vrrp vrid 30 virtual-ip 10.2.2.1
dhcp select global
interface Vlanif40
description DMZ
ip address 10.2.4.3 255.255.255.0
vrrp vrid 40 virtual-ip 10.2.4.1
dhcp select global
interface Vlanif50
description CORE_APPS
ip address 10.2.5.3 255.255.255.0
vrrp vrid 50 virtual-ip 10.2.5.1
dhcp select global
interface Vlanif60
description HR
```



```
ip address 10.2.6.3 255.255.255.0
vrrp vrid 60 virtual-ip 10.2.6.1
dhcp select global
interface Vlanif70
description IT
ip address 10.2.7.3 255.255.255.0
vrrp vrid 70 virtual-ip 10.2.7.1
dhcp select global
dhcp select relay
interface Vlanif80
description VOICE
ip address 10.2.8.3 255.255.255.0
vrrp vrid 80 virtual-ip 10.2.8.1
dhcp select global
interface Vlanif90
description CCTV
ip address 10.2.9.3 255.255.255.0
vrrp vrid 90 virtual-ip 10.2.9.1
dhcp select global
interface Vlanif100
description GUEST
ip address 10.2.10.3 255.255.255.0
vrrp vrid 100 virtual-ip 10.2.10.1
dhcp select global
interface Vlanif110
description ENGINEERS
ip address 10.2.11.3 255.255.255.0
vrrp vrid 110 virtual-ip 10.2.11.1
dhcp select global
interface Vlanif120
ip address 10.2.12.3 255.255.255.0
vrrp vrid 120 virtual-ip 10.2.12.1
interface Vlanif200
description IOT
ip address 10.2.20.3 255.255.255.0
vrrp vrid 200 virtual-ip 10.2.20.1
dhcp select global
interface Vlanif250
description BACKUP
ip address 10.2.21.3 255.255.255.0
vrrp vrid 250 virtual-ip 10.2.21.1
interface MEth0/0/1
interface Eth-Trunk1
description Uplink-to-Core-SW1
port link-type trunk
port trunk allow-pass vlan 2 to 4094
traffic-filter inbound acl 3999
```

```
interface Eth-Trunk2
description Agg-sw2-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
traffic-filter inbound acl 3999
interface Eth-Trunk3
description Agg-sw1-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
traffic-filter inbound acl 3999
interface GigabitEthernet0/0/1
interface GigabitEthernet0/0/2
interface GigabitEthernet0/0/3
interface GigabitEthernet0/0/4
interface GigabitEthernet0/0/5
interface GigabitEthernet0/0/6
interface GigabitEthernet0/0/7
interface GigabitEthernet0/0/8
interface GigabitEthernet0/0/9
interface GigabitEthernet0/0/10
interface GigabitEthernet0/0/11
interface GigabitEthernet0/0/12
interface GigabitEthernet0/0/13
interface GigabitEthernet0/0/14
interface GigabitEthernet0/0/15
interface GigabitEthernet0/0/16
port link-type access
port default vlan 10
interface GigabitEthernet0/0/17
eth-trunk 3
interface GigabitEthernet0/0/18
eth-trunk 3
interface GigabitEthernet0/0/19
eth-trunk 2
interface GigabitEthernet0/0/20
eth-trunk 2
interface GigabitEthernet0/0/21
eth-trunk 1
interface GigabitEthernet0/0/22
eth-trunk 1
interface GigabitEthernet0/0/23
interface GigabitEthernet0/0/24
interface NULL0
ip route-static 0.0.0.0 0.0.0.0 10.2.0.10
ip route-static 10.0.0.0 255.255.0.0 10.2.0.10
ip route-static 10.0.0.0 255.255.0.0 172.16.100.2
ip route-static 172.16.100.0 255.255.255.252 10.2.0.10
```

```
stelnet server enable
user-interface con 0
user-interface vty 0 4
authentication-mode aaa
return
```

10. BR1-core-sw-2

```
sysname BR1-Coreswitch-2
FTP server enable
undo info-center enable
vlan batch 10 20 30 40 50 60 70 80 90 100
vlan batch 110 120 200 250
stp mode rstp
stp instance 0 priority 0
cluster enable
ntdp enable
ndp enable
undo nap slave enable
drop illegal-mac alarm
dhcp enable
diffserv domain default
acl number 3999
rule 5 permit ip source 0.0.0.0 0 destination 255.255.255.255 0
rule 10 permit ip source 10.2.0.0 0.0.0.255 destination 10.2.0.0 0.0.0.255
rule 15 permit ip source 10.2.1.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 20 permit ip source 10.2.2.0 0.0.1.255 destination 10.2.2.0 0.0.1.255
rule 25 permit ip source 10.2.4.0 0.0.0.255 destination 10.2.4.0 0.0.0.255
rule 30 permit ip source 10.2.5.0 0.0.0.255 destination 10.2.5.0 0.0.0.255
rule 35 permit ip source 10.2.6.0 0.0.0.255 destination 10.2.6.0 0.0.0.255
rule 40 permit ip source 10.2.7.0 0.0.0.255 destination 10.2.7.0 0.0.0.255
rule 45 permit ip source 10.2.8.0 0.0.0.255 destination 10.2.8.0 0.0.0.255
rule 50 permit ip source 10.2.9.0 0.0.0.255 destination 10.2.9.0 0.0.0.255
rule 55 permit ip source 10.2.10.0 0.0.0.255 destination 10.2.10.0 0.0.0.255
rule 60 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.11.0 0.0.0.255
rule 65 permit ip source 10.2.12.0 0.0.0.255 destination 10.2.12.0 0.0.0.255
rule 70 permit ip source 10.2.20.0 0.0.0.255 destination 10.2.20.0 0.0.0.255
rule 75 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 80 permit ip source 10.2.0.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 85 permit ip source 10.2.0.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 90 permit ip source 10.2.0.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 95 permit ip source 10.2.2.0 0.0.1.255 destination 10.2.1.0 0.0.0.255
rule 100 permit ip source 10.2.2.0 0.0.1.255 destination 10.2.5.0 0.0.0.255
rule 105 permit ip source 10.2.2.0 0.0.1.255 destination 10.2.21.0 0.0.0.255
rule 110 permit ip source 10.2.4.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 115 permit ip source 10.2.4.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 120 permit ip source 10.2.5.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 125 permit ip source 10.2.5.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
```

```
rule 130 permit ip source 10.2.5.0 0.0.0.255 destination 10.2.6.0 0.0.0.255
rule 135 permit ip source 10.2.6.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 140 permit ip source 10.2.6.0 0.0.0.255 destination 10.2.5.0 0.0.0.255
rule 145 permit ip source 10.2.7.0 0.0.0.255 destination 10.2.0.0 0.0.31.255
rule 150 deny ip source 10.2.7.0 0.0.0.255 destination 10.2.10.0 0.0.0.255
rule 155 deny ip source 10.2.7.0 0.0.0.255 destination 10.2.20.0 0.0.0.255
rule 160 deny ip source 10.2.7.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 165 permit ip source 10.2.8.0 0.0.0.255 destination 10.2.12.0 0.0.0.255
rule 170 permit ip source 10.2.9.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 175 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 180 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 185 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.5.0 0.0.0.255
rule 190 permit ip source 10.2.11.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 195 permit ip source 10.2.12.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 200 permit ip source 10.2.20.0 0.0.0.255 destination 10.2.1.0 0.0.0.255
rule 205 permit ip source 10.2.20.0 0.0.0.255 destination 10.2.21.0 0.0.0.255
rule 210 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.2.0 0.0.1.255
rule 215 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.5.0 0.0.0.255
rule 220 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.6.0 0.0.0.255
rule 225 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.9.0 0.0.0.255
rule 230 permit ip source 10.2.21.0 0.0.0.255 destination 10.2.11.0 0.0.0.255
rule 235 deny ip source 10.2.10.0 0.0.0.255 destination 10.2.0.0 0.0.255.255
rule 240 deny ip source 10.2.20.0 0.0.0.255 destination 10.2.0.0 0.0.255.255
rule 245 deny ip source 10.2.11.0 0.0.0.255 destination 10.2.10.0 0.0.0.255
rule 250 permit ip source 10.2.1.0 0.0.0.255 destination 10.0.1.0 0.0.0.255
rule 255 permit ip source 10.2.2.0 0.0.1.255 destination 10.0.2.0 0.0.1.255
rule 260 permit ip source 10.2.6.0 0.0.0.255 destination 10.0.5.0 0.0.0.255
rule 265 permit ip source 10.2.11.0 0.0.0.255 destination 10.0.11.0 0.0.0.255
rule 270 permit ip source 10.2.21.0 0.0.0.255 destination 10.0.21.0 0.0.0.255
rule 275 deny ip
drop-profile default
ip pool vlan100-sw2
gateway-list 10.2.10.1
network 10.2.10.0 mask 255.255.255.0
excluded-ip-address 10.2.10.251 10.2.10.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan110-sw2
gateway-list 10.2.11.1
network 10.2.11.0 mask 255.255.255.0
excluded-ip-address 10.2.11.251 10.2.11.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan200-sw2
gateway-list 10.2.20.1
network 10.2.20.0 mask 255.255.255.0
excluded-ip-address 10.2.20.251 10.2.20.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan50-sw2
```

```
gateway-list 10.2.5.1
network 10.2.5.0 mask 255.255.255.0
excluded-ip-address 10.2.5.251 10.2.5.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan60-sw2
gateway-list 10.2.6.1
network 10.2.6.0 mask 255.255.255.0
excluded-ip-address 10.2.6.251 10.2.6.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan70-sw2
gateway-list 10.2.7.1
network 10.2.7.0 mask 255.255.255.0
excluded-ip-address 10.2.7.251 10.2.7.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan80-sw2
gateway-list 10.2.8.1
network 10.2.8.0 mask 255.255.255.0
excluded-ip-address 10.2.8.251 10.2.8.254
dns-list 8.8.8.8 1.1.1.1
ip pool vlan90-sw2
gateway-list 10.2.9.1
network 10.2.9.0 mask 255.255.255.0
excluded-ip-address 10.2.9.251 10.2.9.254
dns-list 8.8.8.8 1.1.1.1
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user depi password cipher T70"M%@"'AC=K6%QE'XW;1!!
local-user depi privilege level 10
local-user depi service-type telnet ssh ftp
local-user admin password cipher !<*PMY9[$U)NZPO3J)BXBHA!!
local-user admin privilege level 15
local-user admin service-type telnet ssh ftp
local-user viewer password cipher O.]B_=PL+/J*EP&JX-W=1!!
local-user viewer privilege level 0
local-user viewer service-type telnet ssh ftp
interface Vlanif1
interface Vlanif10
description MGMT
ip address 10.2.0.3 255.255.255.0
vrrp vrid 10 virtual-ip 10.2.0.1
dhcp select global
interface Vlanif20
description INFRA
```

```
ip address 10.2.1.3 255.255.255.0
vrrp vrid 20 virtual-ip 10.2.1.1
dhcp select global
interface Vlanif30
description SERVERS
ip address 10.2.2.3 255.255.254.0
vrrp vrid 30 virtual-ip 10.2.2.1
dhcp select global
interface Vlanif40
description DMZ
ip address 10.2.4.3 255.255.255.0
vrrp vrid 40 virtual-ip 10.2.4.1
dhcp select global
interface Vlanif50
description CORE_APPS
ip address 10.2.5.3 255.255.255.0
vrrp vrid 50 virtual-ip 10.2.5.1
dhcp select global
interface Vlanif60
description HR
ip address 10.2.6.3 255.255.255.0
vrrp vrid 60 virtual-ip 10.2.6.1
dhcp select global
interface Vlanif70
description IT
ip address 10.2.7.3 255.255.255.0
vrrp vrid 70 virtual-ip 10.2.7.1
dhcp select global
dhcp select relay
interface Vlanif80
description VOICE
ip address 10.2.8.3 255.255.255.0
vrrp vrid 80 virtual-ip 10.2.8.1
dhcp select global
interface Vlanif90
description CCTV
ip address 10.2.9.3 255.255.255.0
vrrp vrid 90 virtual-ip 10.2.9.1
dhcp select global
interface Vlanif100
description GUEST
ip address 10.2.10.3 255.255.255.0
vrrp vrid 100 virtual-ip 10.2.10.1
dhcp select global
interface Vlanif110
description ENGINEERS
ip address 10.2.11.3 255.255.255.0
```

```
vrrp vrid 110 virtual-ip 10.2.11.1
dhcp select global
interface Vlanif120
ip address 10.2.12.3 255.255.255.0
vrrp vrid 120 virtual-ip 10.2.12.1
interface Vlanif200
description IOT
ip address 10.2.20.3 255.255.255.0
vrrp vrid 200 virtual-ip 10.2.20.1
dhcp select global
interface Vlanif250
description BACKUP
ip address 10.2.21.3 255.255.255.0
vrrp vrid 250 virtual-ip 10.2.21.1
interface MEth0/0/1
interface Eth-Trunk1
description Uplink-to-Core-SW1
port link-type trunk
port trunk allow-pass vlan 2 to 4094
traffic-filter inbound acl 3999
interface Eth-Trunk2
description Agg-sw2-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
traffic-filter inbound acl 3999
interface Eth-Trunk3
description Agg-sw1-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
traffic-filter inbound acl 3999
interface GigabitEthernet0/0/1
interface GigabitEthernet0/0/2
interface GigabitEthernet0/0/3
interface GigabitEthernet0/0/4
interface GigabitEthernet0/0/5
interface GigabitEthernet0/0/6
interface GigabitEthernet0/0/7
interface GigabitEthernet0/0/8
interface GigabitEthernet0/0/9
interface GigabitEthernet0/0/10
interface GigabitEthernet0/0/11
interface GigabitEthernet0/0/12
interface GigabitEthernet0/0/13
interface GigabitEthernet0/0/14
interface GigabitEthernet0/0/15
interface GigabitEthernet0/0/16
port link-type access
```

```
port default vlan 10
interface GigabitEthernet0/0/17
eth-trunk 3
interface GigabitEthernet0/0/18
eth-trunk 3
interface GigabitEthernet0/0/19
eth-trunk 2
interface GigabitEthernet0/0/20
eth-trunk 2
interface GigabitEthernet0/0/21
eth-trunk 1
interface GigabitEthernet0/0/22
eth-trunk 1
interface GigabitEthernet0/0/23
interface GigabitEthernet0/0/24
interface NULL0
ip route-static 0.0.0.0 0.0.0.0 10.2.0.10
ip route-static 10.0.0.0 255.255.0.0 10.2.0.10
ip route-static 10.0.0.0 255.255.0.0 172.16.100.2
ip route-static 172.16.100.0 255.255.255.252 10.2.0.10
stelnet server enable
user-interface con 0
user-interface vty 0 4
authentication-mode aaa
return
```

11. BR1-Agg-sw-1

```
sysname Agg-SW1
undo info-center enable
vlan batch 10 20 30 40 50 60 70 80 90 100
vlan batch 110 120 200 250
cluster enable
ntdp enable
ndp enable
drop illegal-mac alarm
dhcp enable
dhcp snooping enable
diffserv domain default
drop-profile default
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password simple admin
```



```
local-user admin service-type http
interface Vlanif1
interface MEth0/0/1
interface Eth-Trunk1
description Core-sw1-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
dhcp snooping trusted
interface Eth-Trunk2
description Core-sw2-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
dhcp snooping trusted
interface GigabitEthernet0/0/1
port link-type access
port default vlan 80
interface GigabitEthernet0/0/2
port link-type access
port default vlan 90
interface GigabitEthernet0/0/3
port link-type access
port default vlan 70
interface GigabitEthernet0/0/4
interface GigabitEthernet0/0/5
interface GigabitEthernet0/0/6
interface GigabitEthernet0/0/7
interface GigabitEthernet0/0/8
interface GigabitEthernet0/0/9
interface GigabitEthernet0/0/10
interface GigabitEthernet0/0/11
interface GigabitEthernet0/0/12
interface GigabitEthernet0/0/13
interface GigabitEthernet0/0/14
interface GigabitEthernet0/0/15
interface GigabitEthernet0/0/16
interface GigabitEthernet0/0/17
interface GigabitEthernet0/0/18
interface GigabitEthernet0/0/19
interface GigabitEthernet0/0/20
interface GigabitEthernet0/0/21
eth-trunk 2
interface GigabitEthernet0/0/22
eth-trunk 2
interface GigabitEthernet0/0/23
eth-trunk 1
interface GigabitEthernet0/0/24
eth-trunk 1
```

```
interface NULL0
user-interface con 0
user-interface vty 0 4
port-group link-type
return
```

12. BR1-Agg-sw-2

```
sysname Agg-SW2
undo info-center enable
vlan batch 10 20 30 40 50 60 70 80 90 100
vlan batch 110 120 200 250
stp mode rstp
cluster enable
ntdp enable
ndp enable
drop illegal-mac alarm
dhcp enable
dhcp snooping enable
diffserv domain default
drop-profile default
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password simple admin
local-user admin service-type http
interface Vlanif1
interface MEth0/0/1
interface Eth-Trunk1
description Core-sw1-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
stp edged-port disable
dhcp snooping trusted
interface Eth-Trunk2
description Core-sw2-UPLINK
port link-type trunk
port trunk allow-pass vlan 2 to 4094
dhcp snooping trusted
interface GigabitEthernet0/0/1
interface GigabitEthernet0/0/2
interface GigabitEthernet0/0/3
interface GigabitEthernet0/0/4
port link-type access
```

```
port default vlan 70
interface GigabitEthernet0/0/5
interface GigabitEthernet0/0/6
interface GigabitEthernet0/0/7
interface GigabitEthernet0/0/8
interface GigabitEthernet0/0/9
interface GigabitEthernet0/0/10
interface GigabitEthernet0/0/11
interface GigabitEthernet0/0/12
interface GigabitEthernet0/0/13
interface GigabitEthernet0/0/14
interface GigabitEthernet0/0/15
interface GigabitEthernet0/0/16
port link-type trunk
port trunk allow-pass vlan 100 120
interface GigabitEthernet0/0/17
description File-Storage-UPLINK
port link-type access
port default vlan 20
interface GigabitEthernet0/0/18
description NVR-UPLINK
port link-type access
port default vlan 90
interface GigabitEthernet0/0/19
description VOIP-UPLINK
port link-type access
port default vlan 80
interface GigabitEthernet0/0/20
interface GigabitEthernet0/0/21
eth-trunk 1
interface GigabitEthernet0/0/22
eth-trunk 1
interface GigabitEthernet0/0/23
eth-trunk 2
interface GigabitEthernet0/0/24
eth-trunk 2
interface NULL0
user-interface con 0
user-interface vty 0 4
return
```

13. BR1-AC-1

```
sysname AC-Suez
undo http secure-server enable
set memory-usage threshold 0
ssl renegotiation-rate 1
```

```
vlan batch 100 120
authentication-profile name default_authen_profile
authentication-profile name dot1x_authen_profile
authentication-profile name mac_authen_profile
authentication-profile name portal_authen_profile
authentication-profile name macportal_authen_profile
dhcp enable
dhcp snooping enable
diffserv domain default
radius-server template default
pki realm default
rsa local-key-pair default
enrollment self-signed
ike proposal default
encryption-algorithm aes-256
dh group14
authentication-algorithm sha2-256
authentication-method pre-share
integrity-algorithm hmac-sha2-256
prf hmac-sha2-256
free-rule-template name default_free_rule
portal-access-profile name portal_access_profile
ip pool vlan120
gateway-list 10.2.12.1
network 10.2.12.0 mask 255.255.255.0
dns-list 8.8.8.8 8.8.4.4
aaa
authentication-scheme default
authentication-scheme radius
authentication-mode radius
authorization-scheme default
accounting-scheme default
domain default
authentication-scheme radius
radius-server default
domain default_admin
authentication-scheme default
local-user admin password irreversible-cipher
$1a$7(vSY/OI*,$zcpkA=T\W6Vtum2'.y}O8bDuT].C&$*;,d8@I*5S$
local-user admin service-type http
interface Vlanif120
ip address 10.2.12.252 255.255.255.0
dhcp select global
interface Eth-Trunk1
description AGG-SW3-UPLINK
port link-type trunk
port trunk allow-pass vlan 100 120
```

```
dhcp snooping trusted
interface Eth-Trunk2
description AP-1-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface Eth-Trunk3
description AP-2-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface Eth-Trunk4
description AP-3-UPLINK
port link-type trunk
port trunk pvid vlan 120
port trunk allow-pass vlan 100 120
interface GigabitEthernet0/0/1
eth-trunk 4
interface GigabitEthernet0/0/2
eth-trunk 4
interface GigabitEthernet0/0/3
eth-trunk 3
interface GigabitEthernet0/0/4
eth-trunk 3
interface GigabitEthernet0/0/5
eth-trunk 2
interface GigabitEthernet0/0/6
eth-trunk 2
interface GigabitEthernet0/0/7
undo negotiation auto
duplex half
eth-trunk 1
interface GigabitEthernet0/0/8
undo negotiation auto
duplex half
eth-trunk 1
interface NULL0
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
ssh server secure-algorithms cipher aes256_ctr aes128_ctr
ssh server key-exchange dh_group14_sha1
ssh client secure-algorithms cipher aes256_ctr aes128_ctr
ssh client secure-algorithms hmac sha2_256
ssh client key-exchange dh_group14_sha1
ip route-static 0.0.0.0 0.0.0.0 10.2.12.1
capwap source ip-address 10.2.12.252
user-interface con 0
```

```

user-interface vty 0 4
protocol inbound all
user-interface vty 16 20
protocol inbound all
wlan
traffic-profile name default
security-profile name default
security-profile name default-wds
security-profile name Corelink-Sec
security wpa-wpa2 psk pass-phrase
%^%#;ov#(gw+_h!FeM#1SSI(#DWGrV(o1Uhxm&K!yoT%^%# aes
security-profile name default-mesh
ssid-profile name default
ssid-profile name Corelink-SSID
ssid "Corelink Technologies"
vap-profile name default
vap-profile name Corelink-VAP
service-vlan vlan-id 100
ssid-profile Corelink-SSID
security-profile Corelink-Sec
wds-profile name default
mesh-handover-profile name default
mesh-profile name default
regulatory-domain-profile name default
air-scan-profile name default
rrm-profile name default
radio-2g-profile name default
radio-5g-profile name default
wids-spoof-profile name default
wids-profile name default
wireless-access-specification
ap-system-profile name default
port-link-profile name default
wired-port-profile name default
serial-profile name preset-enjoyor-toeap
ap-group name default
ap-group name CORELINK-APs
radio 0
vap-profile Corelink-VAP wlan 1
radio 1
vap-profile Corelink-VAP wlan 1
ap-id 1 type-id 69 ap-mac 00e0-fcfb-6980 ap-sn 21023544831074426025
ap-name Corelink-AP1
ap-group CORELINK-APs
ap-id 2 type-id 69 ap-mac 00e0-fc41-2440 ap-sn 2102354483109F269208
ap-name Corelink-AP2
ap-group CORELINK-APs

```

```
provision-ap  
dot1x-access-profile name dot1x_access_profile  
mac-access-profile name mac_access_profile  
return
```